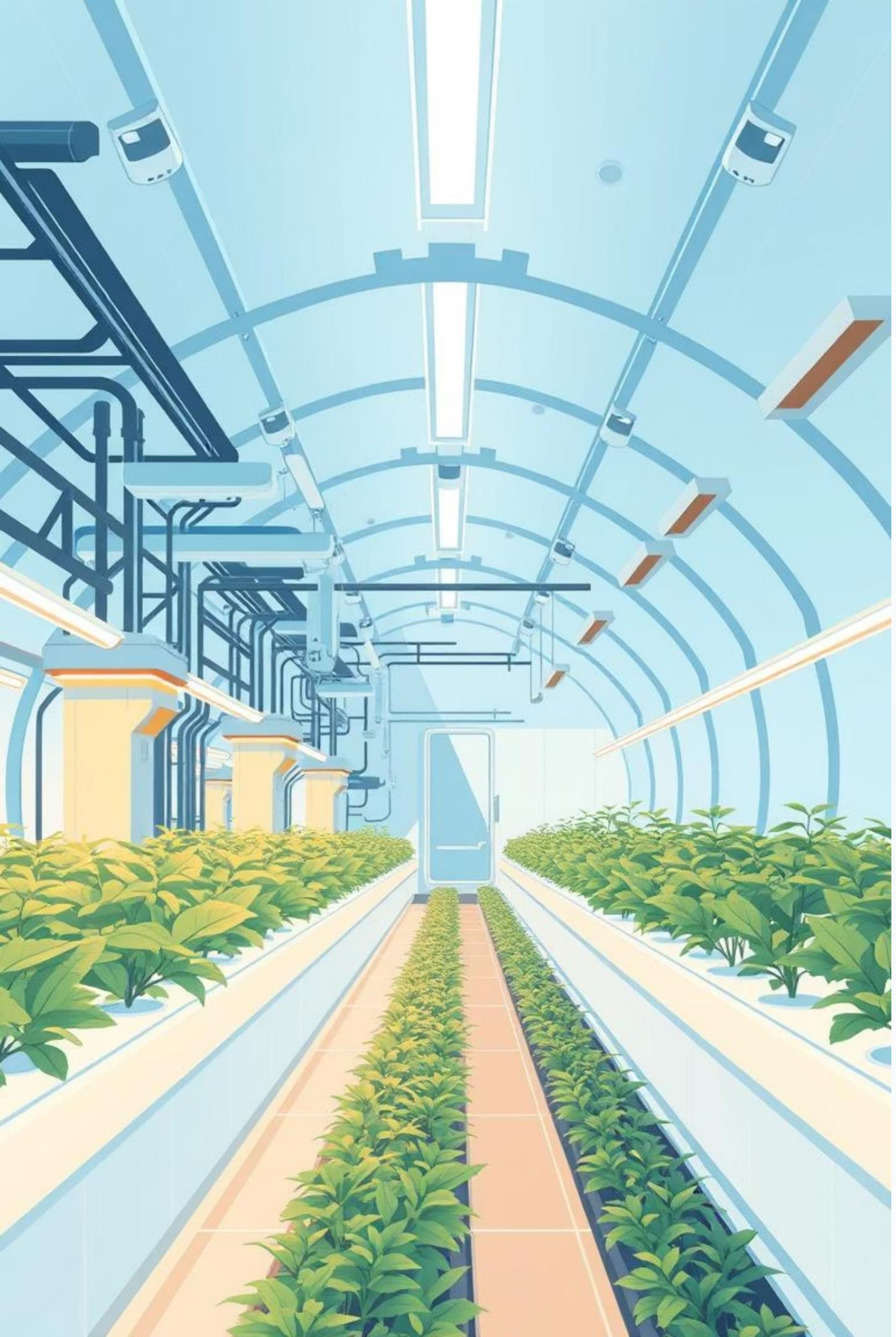


Autonomous Greenhouse Control System

By: Silicon Squad

Advancing agricultural efficiency through intelligent automation.

Introduction

Abstract: Revolutionizing Greenhouse Cultivation

This presentation introduces the **Autonomous Greenhouse Control System**, an embedded solution designed to optimize plant growth conditions through precise environmental management. Developed by Silicon Squad, this system leverages cutting-edge sensor technology, robust microcontroller processing, and intelligent actuation to create a self-regulating greenhouse environment.

The core objective is to move beyond conventional, labor-intensive greenhouse management by automating critical environmental factors such as lighting, irrigation, and ventilation.

Purpose

Enhance crop yield and resource efficiency in controlled agricultural environments.

Components

Integrated sensors (soil moisture, LDR, DHT11), an STM32 microcontroller, and various actuators (water pump, LED grow lights, cooling fan).

Automation

Real-time monitoring and adaptive control of light intensity, soil humidity, and temperature/humidity levels.

Project Contributors

Meet the Silicon Squad

Our multidisciplinary team combines expertise in software, hardware, and integration to deliver robust and innovative embedded solutions.

Rahul Sharma

Lead Developer, responsible for **Coding & Unit Testing**, and **Debugging & Bug Fixing** to ensure system stability.

Kolisetty Jayanth

Project **Scrum Master** and lead for comprehensive **Documentation**, facilitating team coordination and clarity.

Aiswarya

Hardware Specialist, focusing on **Hardware Setup**, **Design** of physical components, and system **Integration**.

Sri Kamesh Guru

Testing Engineer, specializing in system **Integration** and rigorous **Functional Testing** to validate performance.

Karishma

System Architect, overseeing **Architecture Design**, **Hardware Setup**, and overall system **Design** principles.

The Need for Automation

Why Automate Greenhouse Control?

Traditional greenhouse operations often rely on manual monitoring and intervention, which can be inconsistent, labor-intensive, and prone to human error. Even slight deviations in environmental parameters can significantly impact plant health, growth rates, and overall yield.

The primary driver for greenhouse automation is the pursuit of **optimal growing conditions** that maximize output while minimizing resource wastage.

Precision Agriculture

Automation enables granular control over light, water, and temperature, critical for specific plant requirements and accelerated growth cycles.

Resource Optimization

Minimizes water and energy consumption by delivering exact amounts when and where needed, reducing waste and operational costs.

The Challenge

Addressing Inefficiencies in Traditional Greenhouse Management Management

Current greenhouse practices frequently suffer from significant inefficiencies, leading to suboptimal plant growth and considerable resource wastage. Manual control of critical environmental factors is inherently limited in its precision and responsiveness.

These challenges highlight the urgent need for an automated solution that can maintain consistent conditions autonomously.

Manual Control Limitations

Inaccurate and delayed adjustments to environmental parameters such as irrigation, lighting, and ventilation.

Resource Wastage

Over-irrigation, excessive lighting, and inefficient ventilation lead to unnecessary consumption of water and energy.

Inconsistent Growth

Fluctuating environmental conditions result in uneven plant development and reduced crop quality.

Our Solution

Key Objectives of the Autonomous Greenhouse System

Our primary goals are to develop a system that not only automates environmental control but also provides real-time insights and leverages efficient embedded processing for robust and reliable operation.

Automate Core Functions

Implement automated control over essential greenhouse elements: **lighting** for photosynthesis, **irrigation** for soil moisture, and **ventilation** for temperature and humidity regulation.

Monitor Real-Time Parameters

Continuously collect data from crucial sensors, including soil moisture, light intensity (LDR), and ambient temperature/humidity (DHT11), to ensure proactive adjustments.

Efficient Embedded Control

Utilize the **STM32 microcontroller** for its high processing power, low power consumption, and extensive peripheral set, enabling efficient data processing and actuator control.

User-Friendly Interface

Develop an intuitive interface for monitoring system status, viewing sensor data, and setting custom environmental thresholds.

Project Methodology

Software Development Life Cycle (SDLC) Approach

Our project development follows a structured Software Development Life Cycle (SDLC) methodology to ensure systematic progress, comprehensive testing, and a robust final product. This phased approach allows for iterative refinement and thorough validation at each stage.

1

Requirements Gathering

Defining system functionalities, sensor needs, actuator responses, and user expectations.

2

Design & Hardware Integration

Architecting the system, selecting components (STM32, sensors, drivers), and designing circuit diagrams.

3

Development & Implementation

Writing firmware for the STM32, sensor interfacing, and actuator control logic.

4

Testing & Results Analysis

Rigorous testing of sensor accuracy, actuator responsiveness, and overall system stability and efficiency.

5

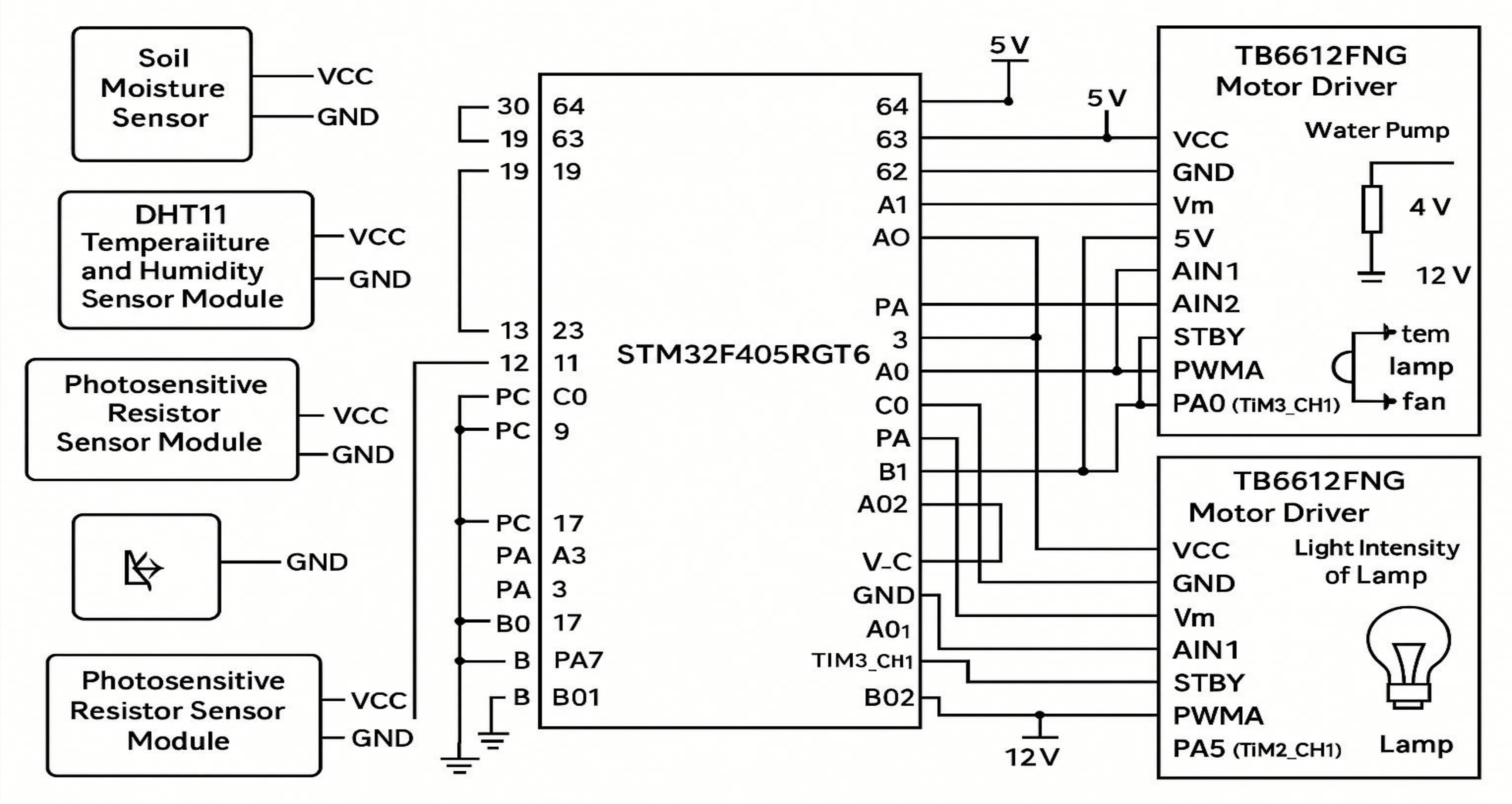
Future Work & Enhancements

Identifying potential upgrades, new features, and scalability options for the system.

Block Diagram: Autonomous Greenhouse Control Flow

The system's architecture is modular, facilitating clear communication pathways and robust control. At its core, the STM32 microcontroller acts as the central processing unit, interpreting sensor data and commanding actuators through dedicated driver circuits.

This structured flow ensures that environmental parameters are continuously monitored, decisions are made efficiently, and actions are executed precisely, leading to optimized greenhouse conditions.



Impact & Adoption

Key Stakeholders and Beneficiaries

The Autonomous Greenhouse Control System is designed to serve a diverse range of users, each benefiting from the system's ability to provide precise, automated environmental management.

Commercial Farmers

Enabling large-scale operations to achieve higher yields, reduce labor costs, and optimize resource use across vast greenhouse areas.

Agricultural Researchers

Providing a controlled and consistent environment for precise experimental conditions, ensuring data integrity and reliable results.

Home Gardeners/Hobbyists

Simplifying greenhouse management, allowing enthusiasts to grow specialized plants with minimal effort and maximized success, even in limited spaces.

Our system caters to both commercial agriculture's scale and individual growers' precision needs.

Operational Logic

Sensor Threshold Justifications

The precise functioning of the autonomous system relies on carefully calibrated thresholds for each sensor. These values are determined based on typical plant physiological requirements and environmental best practices to trigger appropriate actuator responses.

Soil Moisture	< 40% (dry)	When soil moisture drops below 40%, it indicates a need for irrigation to prevent wilting and maintain root health. This threshold ensures optimal hydration without over-watering. The water pump is activated until moisture returns to a healthy range (>50%).
LDR (Light Dependent Resistor)	1800 – 3200 Lux (variable)	This range represents the optimal light intensity for most common greenhouse plants to perform photosynthesis effectively. The system uses a Variable Bulb PWM (Pulse Width Modulation) to modulate the intensity of LED grow lights, ensuring consistent light exposure even during cloudy periods or at night. The exact thresholds can be adjusted for specific crop needs.
Temperature & RH (DHT11)	> 27°C & > 73% RH	Exceeding 27°C can lead to heat stress and reduced growth, while high relative humidity above 73% can promote fungal diseases. If both conditions are met, the cooling fan is activated to reduce temperature and circulate air, thereby lowering humidity and creating a healthier environment for the plants.

These thresholds are crucial for maintaining the delicate balance required for thriving plant life within a controlled greenhouse ecosystem.

Validation

Empirical Support: Insights from Academic Literature

Our system design and chosen thresholds are rigorously supported by recent academic research and established agricultural science. These studies underscore the efficacy of automated environmental control in enhancing yield and resource efficiency.

1

IEEE Access 2021:

A study highlighted that automated irrigation systems, similar to our proposed solution, resulted in an average **30% reduction in water consumption** compared to manual methods, while maintaining or improving crop yield. This validates our soil moisture control strategy.

2

MDPI 2022:

Research demonstrated the existence of specific light intensity thresholds crucial for optimal plant growth across various species. This provides a strong scientific basis for our LDR-based light modulation, aiming to deliver the precise photometric conditions required for maximum photosynthesis and development.

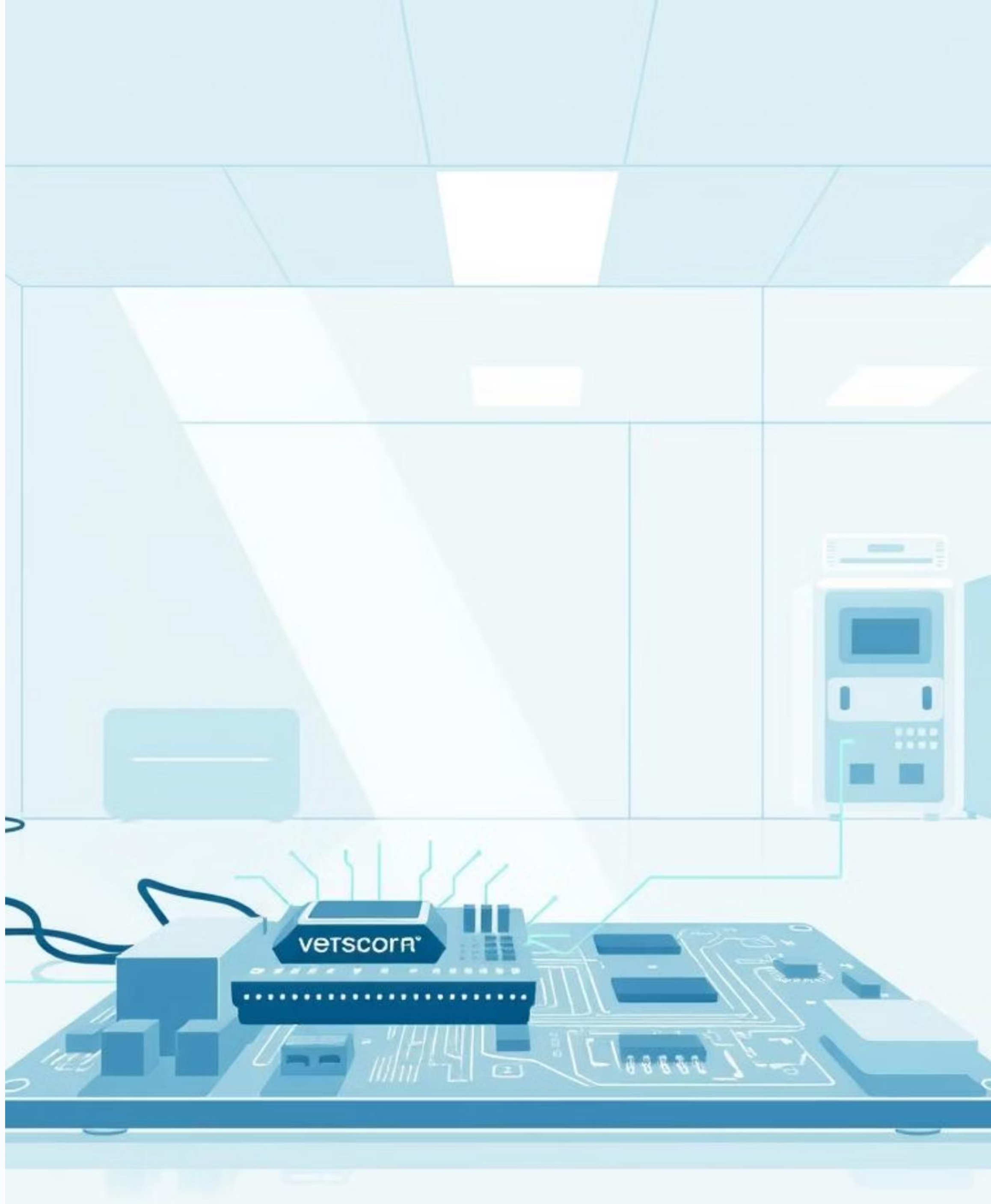
3

Springer IoT 2023:

A comprehensive review of IoT applications in agriculture emphasized the importance of real-time monitoring and automated responses to environmental triggers (temperature, humidity, CO_2) for preventing crop stress and disease. This supports our integrated temperature and humidity control mechanisms.

Sensors & Actuators: Interfacing Principles

A deep dive into the core components that bring intelligent systems to life.



Environmental Monitoring

Soil Moisture Sensing

Understand how we measure soil hydration to optimize irrigation.

Capacitive Principle

Measures changes in the soil's dielectric properties, which vary with moisture content. Unlike resistive sensors, it avoids direct contact with water, preventing corrosion.

Analog Output

Provides a continuous voltage signal. A high voltage output indicates dry soil, while lower voltages correspond to increasing moisture levels.

Durability & Accuracy

Selected for its non-corrosive nature and consistent performance over time, making it ideal for long-term automation in agricultural or smart garden systems.

Soil Moisture Sensor: Interfacing

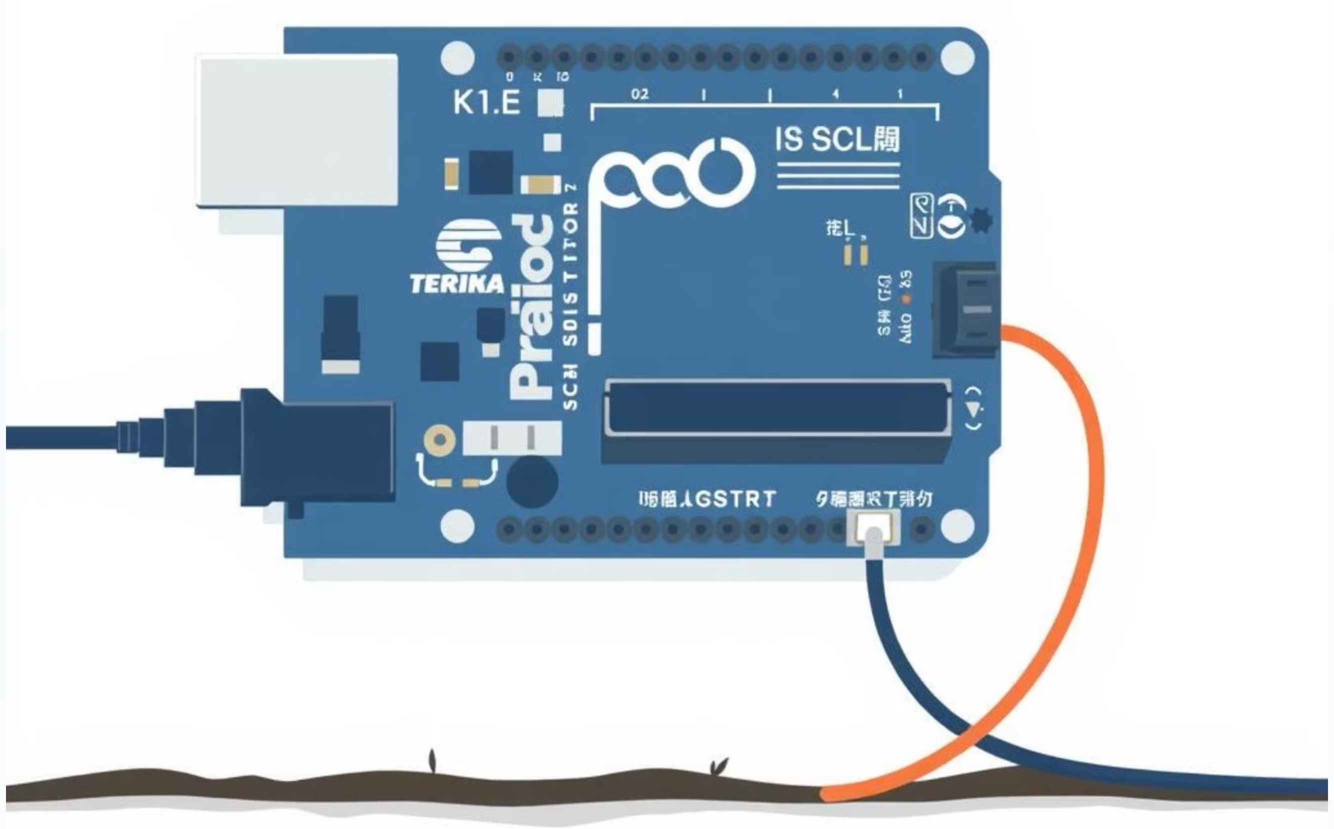
The soil moisture sensor provides an analog output (AO) that needs to be converted into a digital value for processing by the microcontroller.

Interfacing:

- AO (Analog Output) → PA1 (ADC1_IN1)
- Vcc → 3.3V (Power Supply)
- GND → Ground

Threshold Logic:

- If ADC Value > 4050 (representing dry conditions) → Pump ON
- This threshold ensures precise hydration, preventing over or under-watering.



Sensing Ambient Light

Explore the mechanics of light detection for automated lighting control.

Photoresistor Type

An LDR is a cadmium sulfide (CdS) photoresistor whose resistance changes significantly based on the intensity of light falling on it.

Resistance Decreases

As light intensity increases, the LDR's resistance drops. In darkness, its resistance is very high, often in the mega-ohms range.

Analog Output via Voltage Divider

Typically integrated into a voltage divider circuit, converting resistance changes into a readable analog voltage signal for microcontroller input.

Cost-Effective & Reliable

Widely used for basic light sensing due to its low cost, quick response time, and robustness in various environmental conditions.

LDR: Interfacing & Control Logic

The analog output from the LDR's voltage divider circuit is fed into an ADC pin for digital conversion.

Interfacing:

- AO (Analog Output) → PA0 (ADC2_IN0)
- Connect LDR within a voltage divider circuit for stable readings.

Control Logic (for a light source):

- < 1800 (low light): Bulb Off
- 1800–3200 (moderate light): Gradual PWM control, dimming or brightening the bulb proportionately.
- > 3200 (bright light): Bulb at 100% intensity.

DHT11: Temperature & Humidity Sensor

Digital Environmental Monitoring

Discover how we precisely measure ambient temperature and humidity.

Integrated Sensing

The DHT11 combines a Negative Temperature Coefficient (NTC) thermistor for temperature measurement and a capacitive humidity sensor for relative humidity.

Single Digital Output

Unlike analog sensors, the DHT11 provides a pre-calibrated digital signal, simplifying data acquisition and reducing external component count.

Cost-Effective & Efficient

Valued for its low cost, sufficient accuracy (±2°C for temperature, ±5% RH for humidity), and ease of integration into various IoT and environmental monitoring projects.

DHT11: Interfacing & Threshold Logic

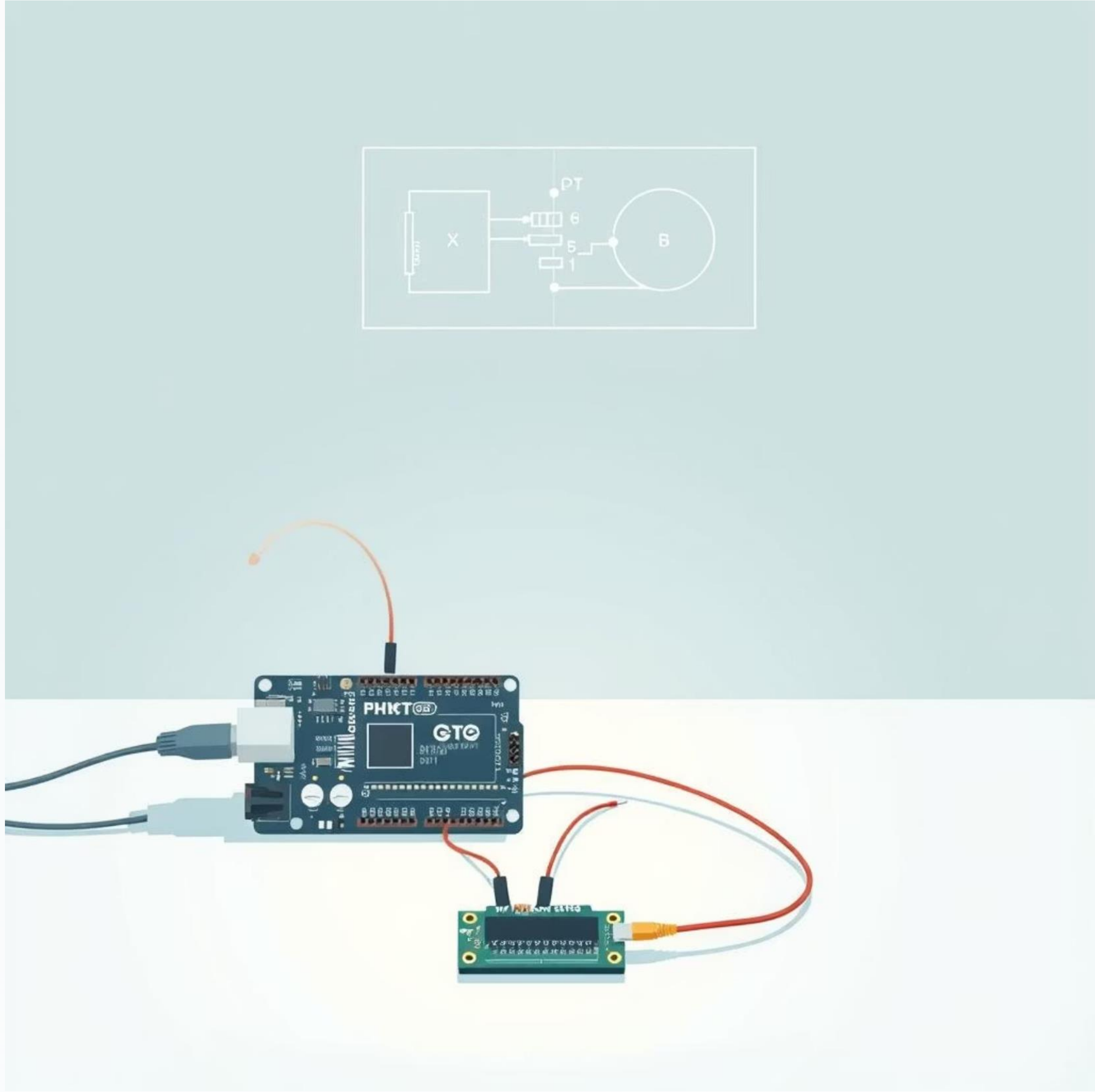
The DHT11 uses a one-wire protocol, making its interfacing straightforward with a single data pin connected to the microcontroller.

Interfacing:

- Data Pin → PA2
- Vcc → 3.3V/5V (check specific module requirements)
- GND → Ground
- A pull-up resistor (typically 4.7kΩ to 10kΩ) on the data line is recommended for stable communication.

Control Logic (for a fan):

- If Temperature > 27°C **AND** Relative Humidity > 73% → Fan ON
- This logic ensures environmental comfort or optimal conditions, preventing heat stress or excessive moisture buildup.



TB6612FNG Motor Driver

Controlling Actuators

Understanding the essential component for precise motor control.

Dual H-Bridge IC

Enables bidirectional control of two DC motors independently, or a single stepper motor, from a low-voltage logic source.

PWM + Logic IN + STBY

Features Pulse Width Modulation (PWM) pins for speed control, logic input pins (IN1, IN2) for direction, and a STBY (standby) pin to enable/disable outputs.

Direction & Speed Control

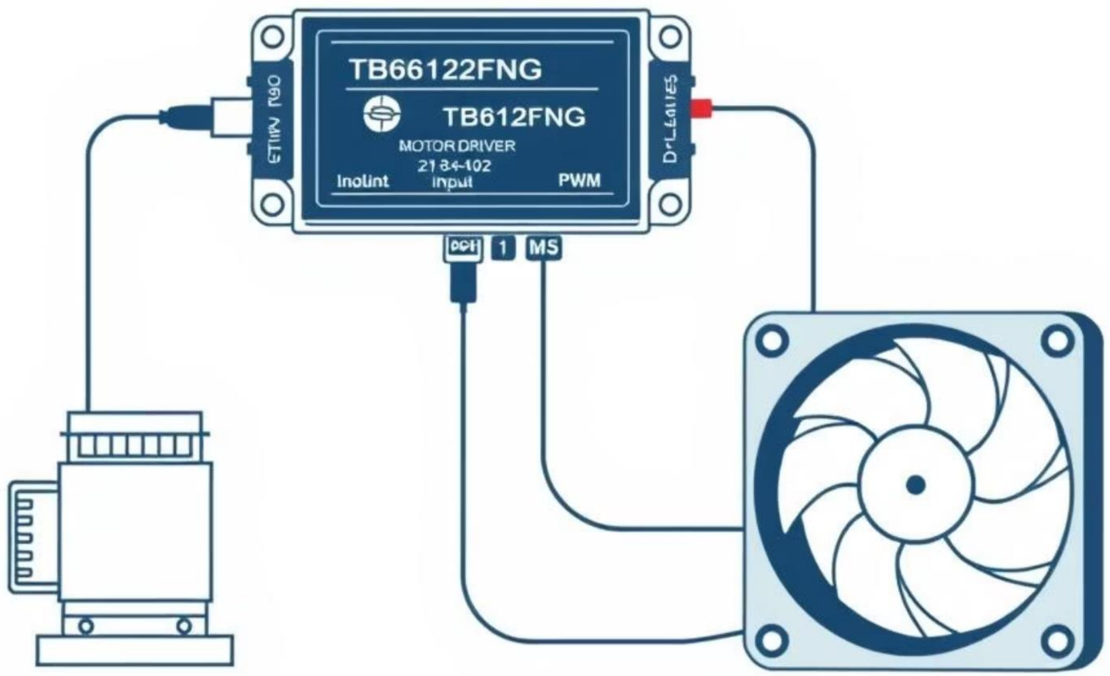
By manipulating the logic inputs and PWM signals, it allows for precise control over both the rotational direction and speed of connected motors.

Motor Driver Wiring: Pump & Fan Control

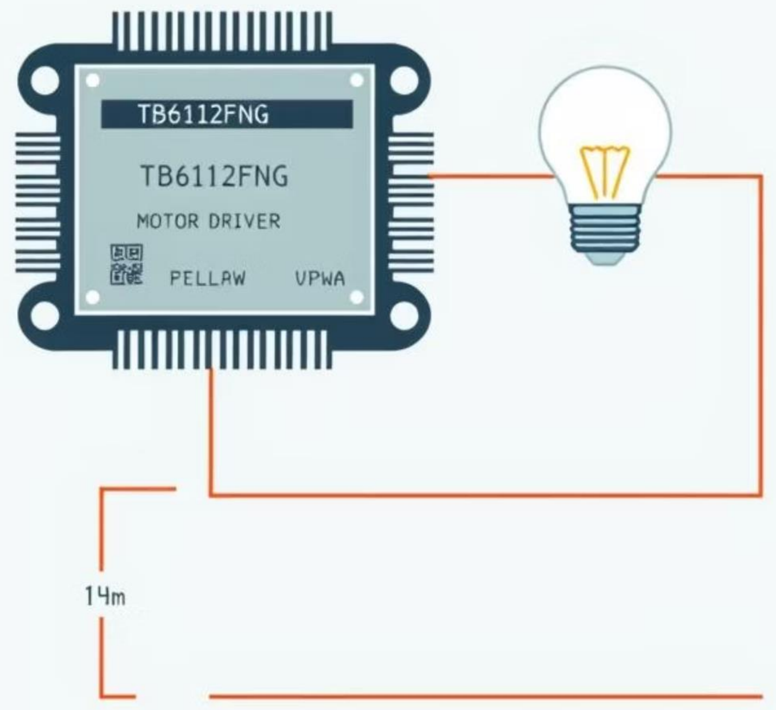
Driver 1 is dedicated to controlling the irrigation pump and the environmental fan, enabling dynamic climate management.

Wiring Details (Driver 1):

- **Pump Control:** AIN1/AIN2 → PC0/PC1; PWMA → PA6
- These connections allow for the pump's direction and speed to be controlled, enabling variable flow rates for irrigation.
- **Fan Control:** BIN1/BIN2 → PC2/PC3; PWMB → PA7
- Similar to the pump, the fan's operation (on/off, speed) can be precisely managed based on environmental readings.
- **Standby:** STBY → PA3
- This pin enables or disables both motor channels on Driver 1, conserving power when motors are not in use.



Motor Driver Wiring: Bulb PWM Control



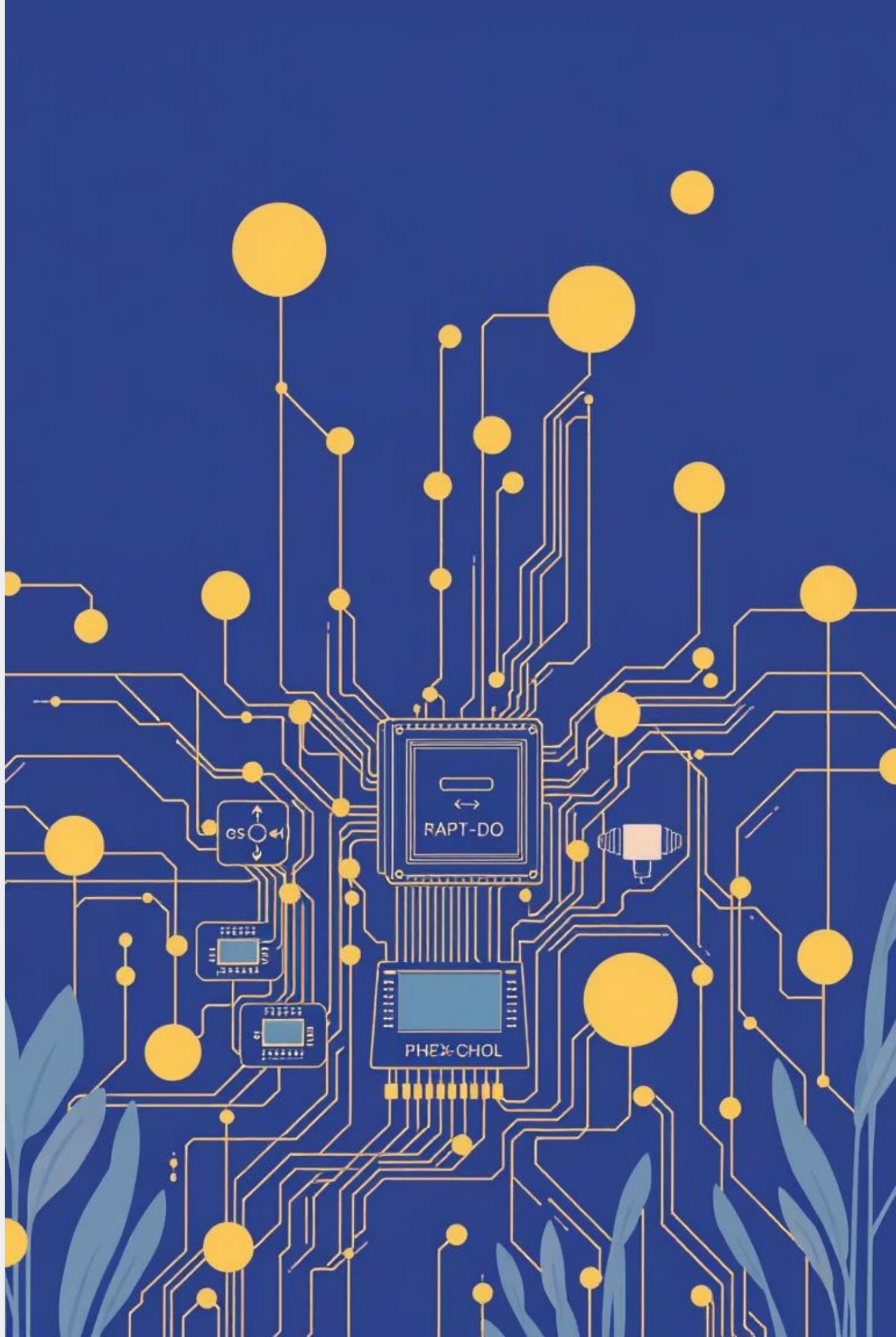
Driver 2 is configured specifically for controlling the illumination lamp, allowing for precise light intensity adjustment using PWM.

Wiring Details (Driver 2 - Lamp):

- **Lamp Control:** AIN1 → PB6; AIN2 → PB7; PWMA → PA5
- By controlling these pins, the lamp's brightness can be modulated via PWM, mimicking natural light cycles or specific illumination requirements.
- **Standby:** STBY → PB5
- This pin controls the entire Driver 2 module, allowing for complete lamp power management.
- **Motor Power:** Vm → 12V for lamp
- Ensure the lamp's power supply matches the motor driver's output capabilities for optimal performance and longevity.

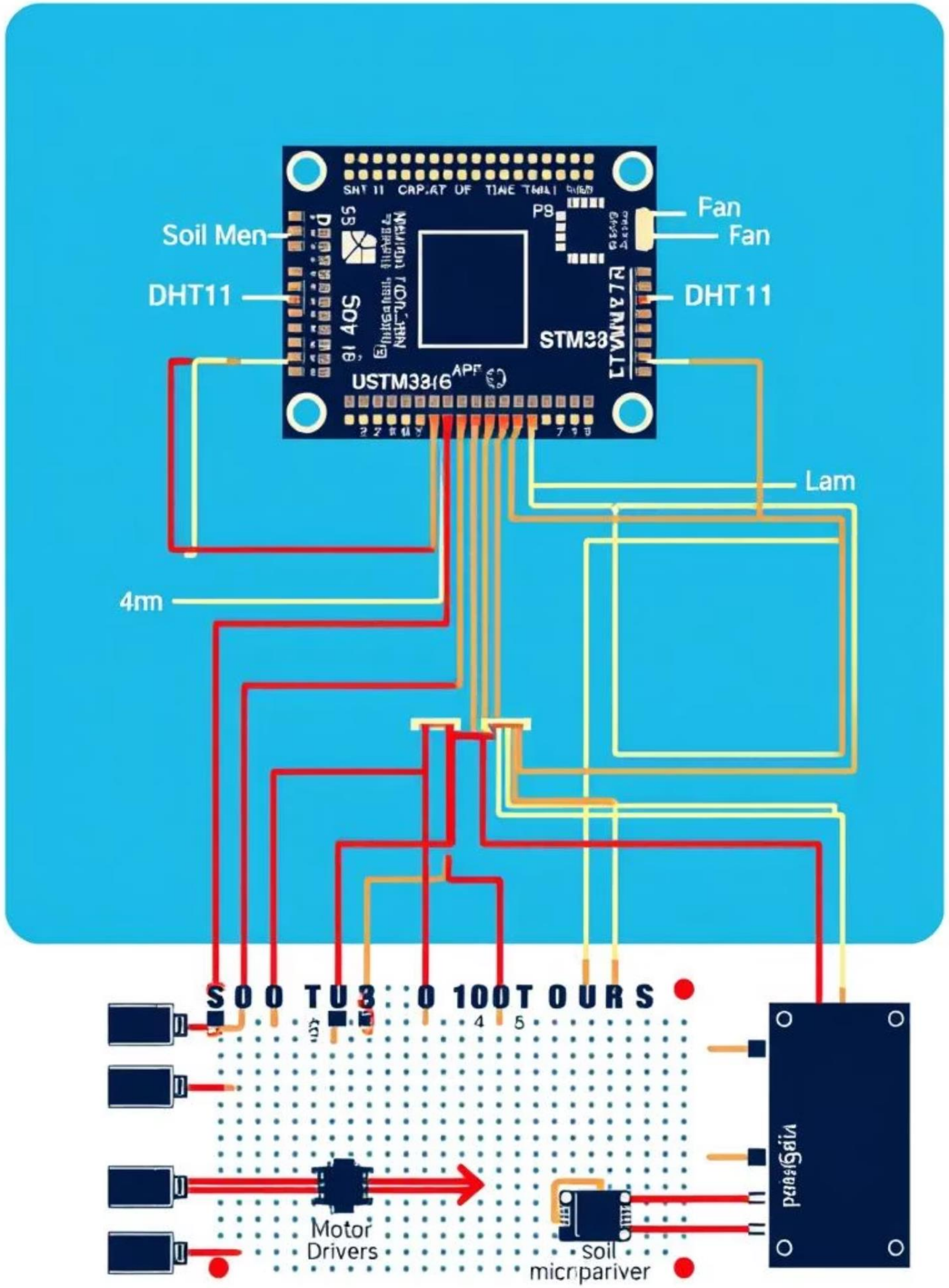
Part 3

Design & Development



Full System Circuit Diagram

The system integrates an STM32 microcontroller with various sensors and actuators to automate environmental control in agriculture. Each component plays a critical role in data acquisition and precise environmental management.



STM32F405RGT6: Core of the System

The STM32F405RGT6 serves as the central processing unit due to its robust specifications and suitability for professional-grade applications.

- **Specs:** 168 MHz CPU, 1 MB Flash, 192 KB SRAM, offering ample processing power and memory for complex algorithms.

Why STM32?

- **Multiple ADCs & PWM Timers:** Essential for precise sensor readings and efficient control of actuators.
- **Faster than Arduino:** Provides the necessary speed for real-time data processing and rapid response.
- **Professional-Grade:** Ensures reliability and scalability for long-term agricultural deployment.

STM32 Pin Mapping Table

The precise pin mapping ensures efficient communication between the STM32 and peripheral components, optimizing system performance.

Component	Pin	Function
Soil Sensor	PA1	ADC1_IN1
LDR	PA0	ADC2_IN0
DHT11	PA2	GPIO
Pump PWM	PA6	TIM3_CH1
Fan PWM	PA7	TIM14_CH1
Lamp PWM	PA5	TIM2_CH1

Embedded Software Logic

The embedded software utilizes STM32CubeIDE's HAL for streamlined development, implementing a continuous loop for real-time environmental control.

1. **HAL in STM32CubeIDE**

   Utilizes Hardware Abstraction Layer (HAL) for simplified peripheral configuration and code portability.

2. **5-Second Loop**

   Main control loop executes every 5 seconds, ensuring timely sensor data acquisition and action.

3. **Sensor Read & Logic**

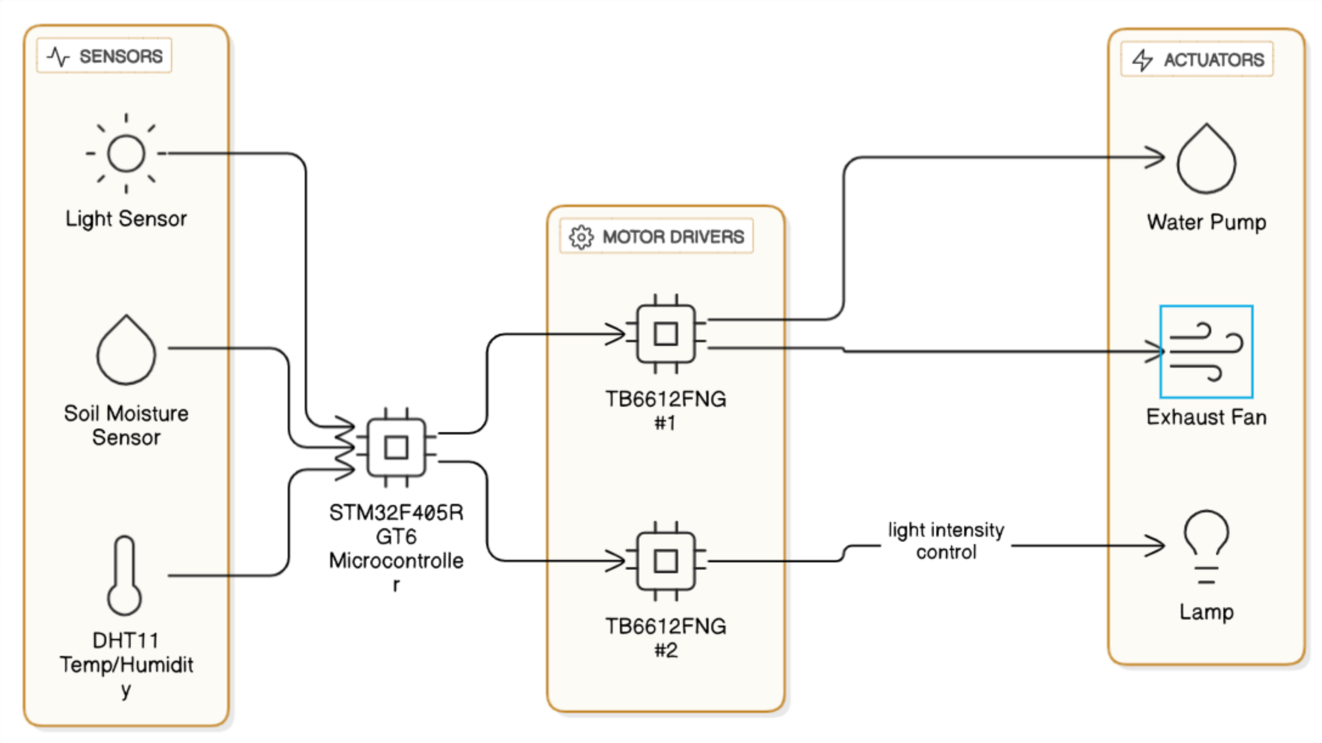
   Reads data from all connected sensors and applies predefined threshold logic to determine necessary actions.

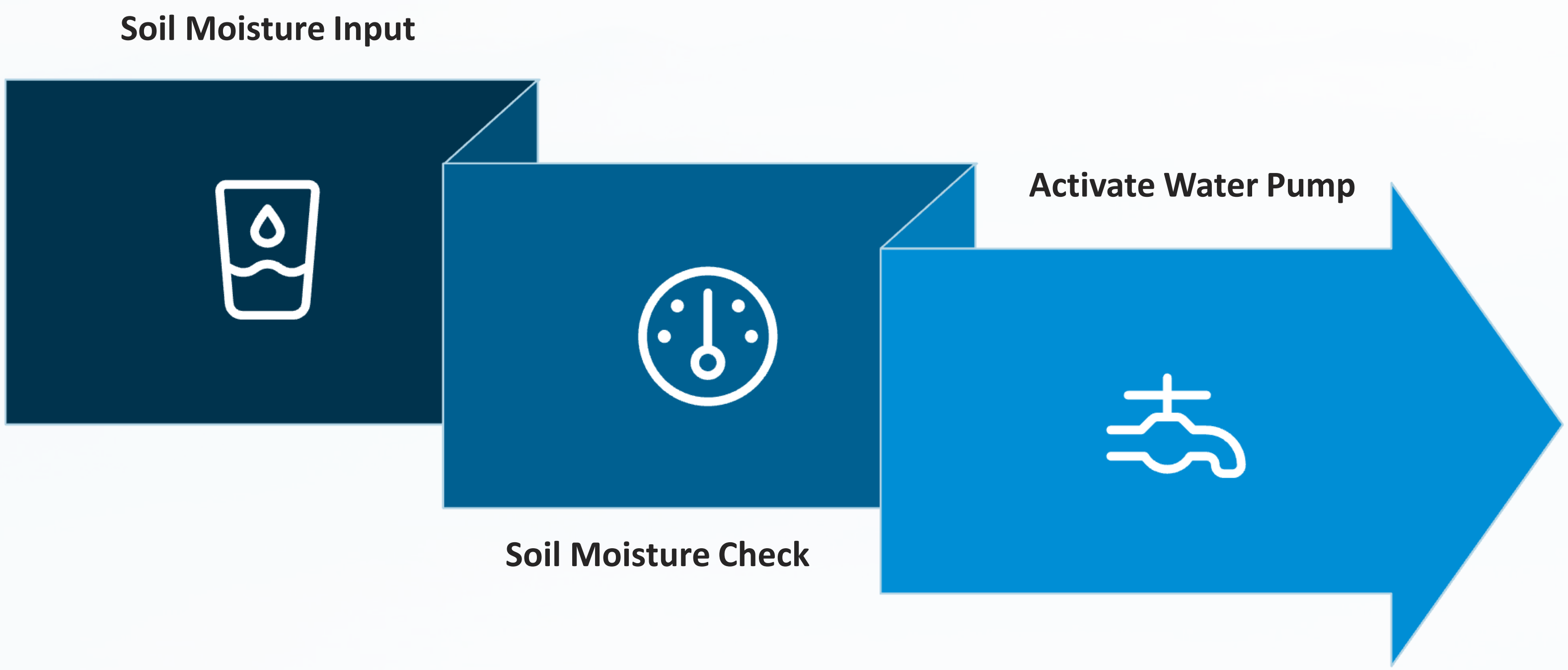
4. **Actuator Control**

   Controls PWM and logic pins for pump, fan, and lamp based on the threshold logic output.

Control Flowchart

The control flowchart visually represents the system's operational logic detailing the input-condition-action sequence for each subsystem.





Part 4

Results, Conclusion & Future Work

Testing & Observations

Initial tests demonstrate the system's responsiveness and effectiveness in maintaining optimal environmental conditions.

Condition	Sensor Output	Action
Dry Soil	4100	Pump ON
Low Light	2500	Bulb PWM 60%
Hot/Humid	29°C/75%	Fan ON

Key Takeaways:

- **Real-time Control:** Achieved immediate response without noticeable delays.
- **Resource Savings:** Observed significant reductions in water and power consumption.
- **Modular Design:** The system's architecture supports easy expansion and adaptation.

Future Enhancements & References

Next Steps for Development

- **Wi-Fi Integration:** Add ESP8266 for remote IoT control.
- **Cloud Data Logging:** Implement data logging to Thingspeak.
- **Adaptive Thresholds:** Incorporate ML for dynamic control.
- **Solar Power:** Transition to sustainable energy for automation.

Key Publications & Resources

- IEEE, MDPI, Springer, Elsevier publications for research insights.
- STM32 and sensor datasheets for technical specifications.
- "Smart Irrigation Using Soil Moisture," IEEE Access, 2021.
- "Light Threshold Control," MDPI Sensors, 2022.

This presentation concludes our detailed overview of the Smart Agriculture System.

THANK YOU